FIBRED TABULATORS

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ABSTRACT. This paper defines tabulators in strict double fibration (i.e., double fibrations of double categories with tabulators). Additionally, we give a Beck-Chevalley Condition for fibred tabulators. Moreover, we show that fibred tabulators are characterized by a preservation property.

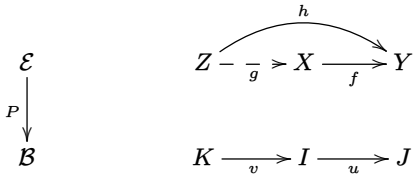
Introduction

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi.

1 Fibrations

In this section we recall some basic facts about fibrations and their examples.

Definition 1.1. Let $P\colon \mathcal{E} \to \mathcal{B}$ be a functor. A morphism $f\colon X \to Y$ in $\mathcal{E}$ is *cartesian* over $u\colon I \to J$ i.e., $P(f) = u$ if for any $v\colon K \to I$ in $\mathcal{B}$ and $h\colon Z \to Y$ over $u \circ v$ there is a unique morphism $g\colon Z \to X$ over v with $h = f \circ g$.



A morphism $k\colon X \to Y$ is called *vertical* if it lies over an identity in $\mathcal{B}$. For an object $J \in \mathcal{B}$, the fibre of P at J denoted by $P(J)$ is the subcategory of $\mathcal{E}$ whose morphisms k lie over id_J.

Definition 1.2. A functor $P\colon \mathcal{E} \to \mathcal{B}$ is called *fibred category* or a *fibration* if for every $Y \in \mathcal{E}$ and $u\colon I \to PY$ there is a cartesian morphism $\bar{u}(E)\colon u^*E \to E$ over u. If P is a fibration, a *cleavage* for P denoted $\varphi(u, E)$ is a choice of cartesian lifts for u at E. Whence P is called a cloven fibration.

1.1 Examples

Let $\mathcal{C}^2$ denote the slice category of $\mathcal{C}$ over itself

Example 1.3 (Fundamental Fibration)**.** Let $\mathcal{C}$ be a category with pullbacks. Then the functor $\mathrm{cod}\colon \mathcal{C}^2 \to \mathcal{C}$ is a fibration; a commutative square is a cartesian arrow in $\mathcal{C}^2$ precisely when it is a pullback.

Example 1.4. The domain functor $\mathrm{dom}\colon \mathcal{C}^2 \to \mathcal{C}$ is a fibration.

Example 1.5 (Family fibration)**.** Every category $\mathcal{C}$ gives rise to a *family fibration* $\mathcal{F}\colon Fam(\mathcal{C}) \to Set$. A morphism $(u, \{\varphi_i\}_{i\in I})$ is cartesian when every φ_i is an isomorphism. Then $\mathcal{F}$ is a fibration since given $u\colon I \to J$ and $\{Y_j\}_{j\in J}$, $(u, \{1_{Y_{u(i)}}\})\colon \{Y_{u(i)}\}_{i\in I} \to \{Y_j\}_{j\in J}$ is cartesian over u.

Date: November 12, 2026.
MSC2020: Primary 00A05, Secondary 00A66.

1.2 Fibred Finite Limits

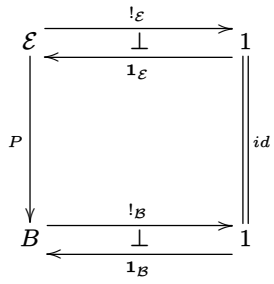
In [Str18] fibred finite limits are given in terms of terminal objects and pullbacks. Furthermore, [Jac99] captures the notion of fibred terminal objects in **Fib** using fibred adjunctions.

Definition 1.6. A fibration $P\colon \mathcal{E} \to \mathcal{B}$ has fibred terminal objects if every fibre $P(I)$ has a terminal object $\mathbf{1}_I$ and they are preserved by cartesianness.

In $\mathbf{Fib}/\mathcal{B}$, the category of fibrations over $\mathcal{B}$, a fibration P has fibred terminals if the fibred functor $F\colon P \to id_B$ has a right adjoint. Moreover, this property is characterized by preservation as follows.

Lemma 1.7. *Let $\mathcal{B}$ be a category with a terminal object $\mathbf{1}_\mathcal{B}$. Then $P\colon \mathcal{E} \to \mathcal{B}$ has fibred terminals if and only if $\mathcal{E}$ has a terminal $1_\mathcal{E}$ and $P(\mathbf{1}_\mathcal{E})$ is a terminal in $\mathcal{B}$.*

Proof. Consider the following diagram in **Fib**



□

Futhermore, we have the following theorem

Theorem 1.8. *If $\mathcal{B}$ has a terminal object $\mathbf{1}_\mathcal{B}$ and $P\colon \mathcal{E} \to \mathcal{B}$ is a fibration, the following are equivalent*

(1) P has a terminal object in $\mathbf{Fib}/\mathcal{B}$.
(2) P has a terminal object in $\mathbf{Fib}$.

Definition 1.9. A fibration $P\colon \mathcal{E} \to \mathcal{B}$ has fibred pullbacks if every fibre has pullbacks and they are preserved by cartesianness.

Similarly, fibred pullbacks can be characterised as follows.

Theorem 1.10. *Let $\mathcal{B}$ be a category with pullbacks. A fibration $P\colon \mathcal{E} \to \mathcal{B}$ has fibred pullbacks if and only if $\mathcal{E}$ has and P preserves pullbacks.*

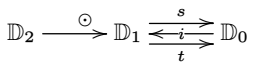
The following result is a consequence of Lemma 1.7 and Theorem 1.10

Theorem 1.11. *For a category $\mathcal{B}$ with finite limits, a fibration $P\colon \mathcal{E} \to \mathcal{B}$ has fibred finite limits if and only if $\mathcal{E}$ has finite limits and P preserves finite limits.*

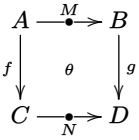
2 Double Categories

2.1 Basic Notions

Definition 2.1. A double category $\mathbb{D}$ is a pseudo category in **Cat**, it consists of the following diagram



We call $\mathbb{D}_0$ the category of objects and vertical arrows and $\mathbb{D}_1$ the category of proarrows and cells:



We refer to M and N as the internal domain and codomain of θ respectively, whilst f and g are the external source and target respectively. Cells with proarrow (resp. vertical) identities will be called *horizontal* (resp. *vertical*) *globular* cells. Composition is strict in the vertical direction and weaker in the horizontal direction. For convention, we assume our double categories to be pseudo in the sense of [CLPS22].

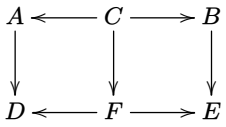
Definition 2.2. Given a double category $\mathbb{D}$, we can form a *horizontal bicategory* $\mathcal{H}(\mathbb{D})$, whose objects are the objects of $\mathbb{D}$, 1-cells are the proarrows of $\mathbb{D}$, and 2-cells are given by vertical globular cells. Dually, we can form a *vertical 2-category* $\mathcal{V}(\mathbb{D})$ whose objects are the objects of $\mathbb{D}$, 1-cells are the vertical arrows, and 2-cells given by horizontal globular cells.

Notation 2.3. For cells α and β, we shall write $\frac{\alpha}{\beta}$ as their vertical composition.

2.2 Examples

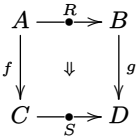
Example 2.4. Let $\mathbb{D}^2$ be the slice double category of $\mathbb{D}$ over itself with external source and target operations coming from $\mathbb{D}$.

Example 2.5. Let $\mathcal{C}$ be a category with pullbacks, we define a double category $\mathbb{S}\mathbf{pan}(\mathcal{C})$ whose vertical category is $\mathcal{C}$, whose proarrows $A \nrightarrow B$ are spans $A \xleftarrow{f} C \xrightarrow{g} B$ in $\mathcal{C}$, and whose cells are commuting squares:



in $\mathcal{C}$. Horizontal composition is given by pullbacks.

Example 2.6. The double category $\mathbb{R}\mathrm{el}(\mathrm{Set})$ has objects as sets and vertical arrows as set-functions. The proarrows are given by relations and there is a unique cell



whenever $R(a,b) \implies S(f(a),g(b))$ holds for all $a \in A, b \in B$.

Definition 2.7. Let $\mathbb{E}$ and $\mathbb{B}$ be pseudo categories internal to a 2-category $\mathcal{K}$ with 2-pullbacks. A *lax functor* $L\colon \mathbb{E} \to \mathbb{B}$ consists of arrows $L_0\colon E_0 \to D_0$ and $L_1\colon E_1 \to B_1$ together with comparison morphisms given by



such that the source and target are strictly preserving. Moreover, ϕ, ι, and the associators and unitors of $\mathbb{E}$ and $\mathbb{B}$ satisfy the usual coherence conditions as in [MF06]. A lax functor is called **strict** if ϕ and ι are identities.

Remark 2.8. A strict functor in **Cat** corresponds to a strict functor between double categories.

We recall the notion of double limits i.e., limits in double categories introduced in [GP99]. However, our limits will be analogous to limits of Grandis and Paré.

Definition 2.9 ([DPP10])**.** A double category $\mathbb{E}$ has *vertical pullbacks* if the diagram

$$\mathbb{E}_2 \xrightarrow{\odot} \mathbb{E}_1 \underset{t}{\overset{s}{\underset{\longleftarrow i}{\rightrightarrows}}} \mathbb{E}_0$$

lies in **PBCat** the category of categories with pullbacks and pullback-preserving functors.

A functor $G\colon \mathbb{E} \to \mathbb{A}$ between double categories with vertical pullbacks preserves pullbacks if it preserves pullbacks componentwise. When this is the case, we shall call G a pullback-preserving functor. For the rest of the paper, we shall refer to vertical pullbacks by v-pullbacks.

Remark 2.10. The universal property of v-pullbacks in double categories generalizes the universal property of 2-pullbacks. $\odot$ preserving pullbacks amounts to a functorial choice of pullbacks, see [GP99].

3 Double Fibrations

Lemma 3.1 ([CLPS22])**.** *Given a diagram $s\colon P_1 \to P_0 \leftarrow P_1 \colon t$ in* **cFib***, a pseudo category structure in* **Fib** *(resp.* **cFib***) is the same as two pseudo double category structure obtained by applying the top and bottom 2-functors,*

Definition 3.2 ([CLPS22])**.** A strict double functor $P\colon \mathbb{E} \to \mathbb{B}$ between pseudo double categories given by

$$\begin{array}{ccccc}
\mathcal{E}_2 & \xrightarrow{\odot_{\mathcal{E}}} & \mathcal{E}_1 & \underset{t_{\mathcal{E}}}{\overset{s_{\mathcal{E}}}{\underset{\longleftarrow i_{\mathcal{E}}}{\rightrightarrows}}} & \mathcal{E}_0 \\
\downarrow{\scriptstyle \mathcal{P}_2} & & \downarrow{\scriptstyle \mathcal{P}_1} & & \downarrow{\scriptstyle \mathcal{P}_0} \\
\mathcal{B}_2 & \xrightarrow[\odot_{\mathcal{B}}]{} & \mathcal{B}_1 & \underset{t_{\mathcal{B}}}{\overset{s_{\mathcal{B}}}{\underset{\longleftarrow i_{\mathcal{B}}}{\rightrightarrows}}} & \mathcal{B}_0
\end{array}$$

defines a **double fibration** when P_0 and P_1 are cloven fibrations, such that $s_{\mathcal{E}}$ and $t_{\mathcal{B}}$ preserve cleavages simultaneously, and $\odot_{\mathcal{E}}$ and $i_{\mathcal{E}}$ preserve cartesian morphisms.

Remark 3.3. A double fibration $P\colon \mathbb{E} \to \mathbb{B}$ is a pseudo category internal to **Fib** such that the source and target maps preserve chosen cleavages. A cleavage for P is chosen componentwise. Since $s_{\mathcal{E}}$ and $t_{\mathcal{B}}$ preserve cleavages, they are morphisms in **cFib**.

References

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